

Clearer Lub-Dub: A Novel Approach in Heart Sound Denoising Based on Transfer Learning

Peng Gao^{1,2†}, Haojie Zhang^{1,2†}, Lin Shen^{1,2}, Yongxin Zhang³, Jiang Liu⁴, Kun Qian^{1,2*},
Bin Hu^{1,2*}, Björn W. Schuller^{5,6}, and Yoshiharu Yamamoto⁷

1. Key Laboratory of Brain Health Intelligent Evaluation and Intervention,
(Beijing Institute of Technology), Ministry of Education, Beijing, China

2. School of Medical Technology, Beijing Institute of Technology, Beijing, China

3. Graduate School of Engineering Science, Osaka University, Osaka, Japan

4. Global Center for Science and Engineering, Faculty of Science and Engineering,
Waseda University, Tokyo 169-8050, Japan

5. GLAM – the Group on Language, Audio, & Music, Imperial College London, UK

6. CHI – Chair of Health Informatics, MRI, Technical University of Munich, Germany

7. Educational Physiology Laboratory, Graduate School of Education, The University of Tokyo, Tokyo, Japan

Corresponding authors: {qian, bh}@bit.edu.cn

Abstract—Cardiovascular diseases (CVDs) constitute the primary cause of human mortality globally in recent decades. To effectively detect CVDs, heart auscultation plays an important role in early diagnosis. With the development of artificial intelligence (AI), many studies have designed varying AI-assisted diagnosis systems helping people discriminate abnormal heart sounds. Yet, a robust system usually requires a noise-less input signal, which is critical as heart sounds are often affected by some unavoidable noise. Therefore, many heart sound classification models use filters or other methods to obtain the clean signals. However, these classic techniques are not adaptable enough to distinguish the meaningful murmurs and real noises. Thus, we propose a novel approach to transfer an audio source separation model to denoise the heart sound. In this paper, we test different denoisers on synthesis heart sound with additive white Gaussian noises. Our method performs well on the noise reduction metrics. Meanwhile, we evaluate the classification performance of each denoiser with some classifiers on the PhysioNet dataset. Experimental results demonstrate that our method can outperform other denoising techniques by achieving the highest unweighted average recall (UAR) at 95.7% with the smallest standard deviation. The results confirm that our method is robust and adaptable in improving audio's denoising.

Index Terms—Heart Sounds, Denoising, Audio Source Separation, Signal Processing, Transfer Learning

I. INTRODUCTION

The strong and rhythmic beat of the heart pumps blood flowing to all parts of the body and gives rise to our lives. This powerful strength associated with mechanical vibrations can be heard and recorded as the heart sound. On account

This work was partially supported by the National Natural Science Foundation of China (Nos. 62272044, and 62227807), the JST Moonshot R&D Program (No. JPMJMS229B), the National Key R&D Program of China (No. 2023YFC2506804), the Ministry of Science and Technology of the People's Republic of China with the STI2030-Major Projects (Nos. 2021ZD0201900, and 2021ZD0200601), and the Teli Young Fellow Program from the Beijing Institute of Technology, China. (Corresponding authors: Kun Qian and Bin Hu.)

Peng Gao and Haojie Zhang contributed equally to this work.

of its noninvasive and easy acquisition with a simple stethoscope, the heart sound is one of the most routinely acquired physiological signals in clinical practice. It provides crucial information for physicians, indicating whether a patient has cardiovascular diseases (CVDs) [1]. Nowadays, CVDs are the leading cause of death worldwide. Reported from World Health Organization's (WHO) fact sheets, an estimated 17.9 million people died from CVDs in 2019, representing 32% of all global deaths [2]. Cardiac auscultation serves as the first whistleblower in detecting heart diseases, which relies on the heart sound evaluation. Computer-aided diagnostic systems play an important role in heart sound auscultation [3]. However, one aspect requiring careful consideration is that these systems rely on a high-quality input signal.

The task of heart sound classification mainly depends on the graphical representation of the waveform of cardiac sounds which is named as phonocardiograph (PCG). Normally, there are two peaks with two quiet intervals in a heart sound cycle, referred to as the period of systole and diastole. When heart damage occurs, there will be a turbulent blood flow in the interval which is called a murmur [4]. But due to its low amplitude, a PCG record is quite susceptible to noise. Unfortunately, there are always inevitable noise sources in clinical practice, such as instrument fricatives, movement artifacts and physiological noises from the lungs and the stomach [5]. Therefore, effective denoising is a necessary preprocessing for AI-assisted diagnosis.

Previous works about heart sound denoising are almost all found in the conventional signal processing field [6] by expert-based approaches to a great extent. Wavelet transformation and bandpass filtering are the commonly used approaches to denoise PCG signals, but they have some delicate problems, respectively [7]. In terms of wavelet transformation, aside from the fact that the selection of the wavelet family needs to rely on experience and practice, it is also a challenge to determine

a good threshold for the coefficients. And as for the bandpass filters, this can be discussed in two categories: the FIR filter and the IIR filter. An FIR filter is merely like a convolution in the time domain, which averages the neighbouring signals with designed weights. It can preserve the signal waveform, but always needs far more coefficients to achieve the desired attenuation, which means higher costs. An IIR filter used in PCG denoising is usually the Butterworth filter which possesses a flatten passband, while its cutoff frequency is hard to select to be the optimum [8]. This is because of both the spectral overlap between noises and murmurs and the gentle roll-off of the Butterworth filter, which makes it either eliminate the essential murmurs or retain the noises.

Overall, we argue that the drawbacks discussed above are caused by the inability of these methods to adapt to the varying signals. With deep neural networks rising up, we can leverage their power to help us go through these difficulties. In fact, there have been already proposed lots of deep learning based models to enhance audio clarity, especially for human speech. Wave-U-Net [9] is one of the classical architecture in end-to-end audio source separation, which works directly on the raw audio waveform and combines multiple scales of abstraction and time resolution of features. Till now, Wave-U-Net has been implemented and modified in many old and new works, presenting its well-designed structure [10], [11].

Motivated by the successful approach of using music source separation architectures in speech enhancement [12], we adopt a state-of-the-art (SOTA) music source separation model to implement our heart sound denoising model. The foundation model is named Hybrid Transformer Demucs (HT Demucs) [13] which is based on a bi-U-Net architecture inspired by the Wave-U-Net and has a cross transformer layer inside. On account of the lack of clean heart sound data, we use transfer learning techniques to fine-tune a pretrained HT Demucs model, adapted from the drums domain to the heart bumps domain.

In this paper, we introduce the HT Demucs model to heart sound denoising and classification tasks for the first time. Our proposed model learns audio features in both the time domain and spectrogram domain. In the transfer learning process, we use Low-Rank Adaptation (LoRA) as method [14] to fine-tune the pretrained HT Demucs on heart sounds and validate it with the UAR of the classification. Then, we design a contrast experiment to compare our approach with classic denoising methods on denoising indicators. In addition, we build a heart sound diagnostic evaluation system to demonstrate the medical contribution of our noise reduction method to the diagnosis by comparing the accuracy in identifying CVDs.

II. MATERIALS AND METHODS

A. Lub-Dub

In this section, we implement Hybrid Transformer Demucs (HT Demucs) and fine-tune it using LoRA method to improve training efficiency.

a) *Hybrid Transformer Demucs*: The HT Demucs model owns a double U-Net encoder/decoder structure, with one in the time domain and another in the spectrogram domain. Four encoder layers and four decoder layers composite each U-Net, and a cross-domain transformer encoder is used to connect and combine the features of both sides. It is an end-to-end architecture, which means that we can just input a raw audio waveform data, and get a denoised audio back. HT Demucs outperforms most similar models in Signal-to-Distortion Ratio (SDR) with less extra training data [13]. We adopt the HT Demucs model pretrained on the MUSDB18-HQ dataset, which is the basic model that [13] published. On account of the similarity of auditory sensational difference between drums and heart bumps, we make an approach to transferring the drum source channel's output to the denoised heart sound audio.

b) *LoRA*: Although HT Demucs has a strong representation ability, it is well-known that the Transformer-based model has been always hard to train or fine-tune. Therefore, we implement LoRA, an efficient fine-tuning technique, which freezes the pretrained model weights and injects trainable rank decomposition matrices into each layer of the Transformer architecture. This technique greatly reduces the number of trainable parameters for downstream tasks, and minimises degradation of model performance in the mean time. In our implementation, we only inject trainable parameters into the linear layers to highly accelerate our training progress.

B. CVDs Diagnosis

Cardiac disease diagnosis is our object, where we train and evaluate our denoising method. Thus, we implement all classification periods prepared for the next stage of the experiment. As shown in Fig. 1, a general AI-aided diagnosis flow contains preprocessing (denoising and segmentation), feature extraction, and classification. We explain these components below.

a) *Segmentation and Feature Extraction*: Here, heart sound segmentation's main idea is to make all the heart sounds be of the same length and have an entire cardiac cycle simultaneously. This also has a good side effect that such segmentation naturally augments the dataset. We use a trained Hidden Semi-Markov Model with logistic regression to recognise each state of heart sounds and then segment them according to the method described in [15]. Specifically, the heart sound is segmented at each beginning of the S1 period and lasts for 2.5 seconds, with zero-padding applied for which is shorter [16].

After segmentation, we extract the time-frequency feature of heart sounds using the Log Mel spectrogram. Mel-scale features are widely used in audio classification [6], [17], [18], because human's audition senses frequency logarithmically. To obtain a Log Mel spectrogram, we first calculating short-time Fourier transform (STFT) with a frame length of 80, 1024 FFT points, and the Hann window function. Then, we multiply 64 Mel filter banks to the spectrogram obtained above and finally get the Log Mel spectrogram by applying the logarithm.

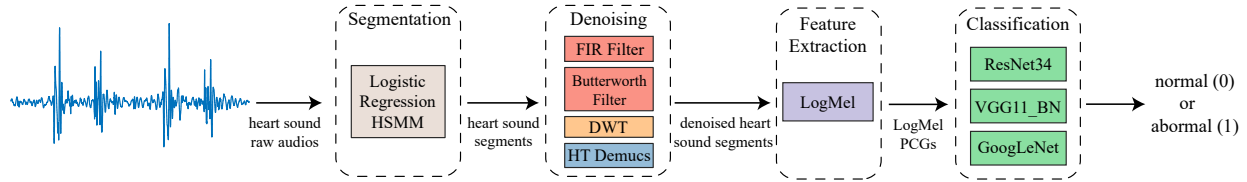


Fig. 1. An overall pipeline structure for our heart sound classification experiments.

A demonstration of the raw waveform and Log Mel feature map is shown in Fig. 2.

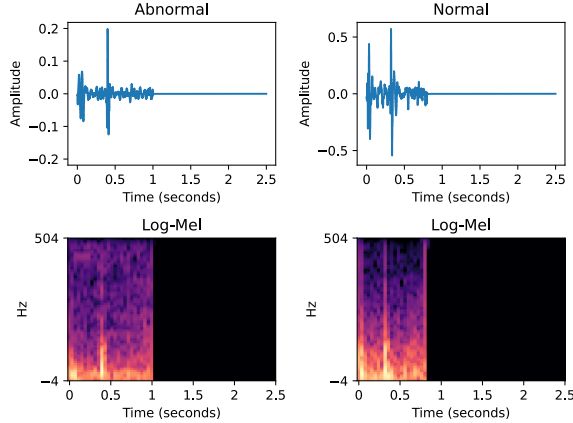


Fig. 2. Waveform (top), and Log Mel spectrogram (bottom) of the heart sound segment samples (in 2.5 s) annotated as ‘abnormal’ (left), and ‘normal’ (right).

b) Pretrained Classifier: In order to evaluate the effect of our denoiser, we first implement pretrained heart sound classifiers using GoogLeNet [19], VGG [20], and ResNet [21]. They all have achieved superior performance becoming among the classic CNN architecture nowadays. In addition, they are also proven to be fairly qualified at SOTA level on heart sound classification task [17], [18], [22]. Specifically, we select the 34-layer ResNet (denoted as ResNet34) and the 11-layer VGG with a batch norm layer after each convolution layer (denoted as VGG11_BN). Here, we modify the first and the last layer of the models to adapt our required data shape. Then, they will be fully fine-tuned on the heart sounds with each denoising method respectively.

TABLE I
EXPERIMENT DATASET AND SEGMENTS.

Subset name	# Raw Recordings			# Segmented Recordings		
	Abnormal	Normal	Total	Abnormal	Normal	Total
Training-a	292	117	409	9 864	4 303	14 167
Training-b	104	386	490	589	2 396	2 985
Training-c	24	7	31	1 425	356	1 781
Training-d	28	27	55	493	308	801
Training-e	183	1 958	2 141	2 852	54 815	57 667
Training-f	34	80	114	1 140	3 012	4 152
Total	665	2 575	3 240	16 363	65 190	81 553

III. EXPERIMENTS AND RESULTS

A. Dataset

In this section, we use the PhysioNet/CinC Challenge 2016 dataset to train and test the proposed method [23]. A total of 3 240 heart sound recordings gathered from seven independent research teams are categorised as normal and abnormal by expert labelling. Specifically, 2 575 heart sounds collected from healthy subjects are labelled as normal, and 665 heart sounds from pathological patients are labelled as abnormal. Table I’s Raw Recordings part shows a brief information of the data distribution. In addition, all the recordings lasting from 5 s to just over 120 s are resampled to 2 kHz, and provided as ‘.wav’ format. After segmentation, we collect a total of 81 553 segments eventually. A detailed distribution is shown in Table I’s Segmented Recordings part.

B. Denoising Methods Used for Comparison

We contrast another 3 classic denoising methods in the same train and test conditions for comparing with our proposed method, which are discrete wavelet transform (DWT), FIR filter, and Butterworth filter. Ali *et al.* [24] concluded that the Db10 wavelet with 4th level of decomposition can give the maximum Signal-to-Noise ratio (SNR) and the minimum standard error (RMSE) of the standard heart sounds. Thus, we follow their DWT configuration and apply a level-dependent threshold with the soft thresholding function to suppress the noise. The level-dependent threshold (τ) is defined as [25]:

$$\tau_j = \sigma_j(\sqrt{2 \ln N_j}) \quad (1)$$

$$\sigma_j = \frac{\text{MAD}(D_j)}{0.6745}, \quad (2)$$

where MAD represents the the median absolute deviation of the detailed coefficients for each level (D_j) and N_j is the number of the samples for each level. Besides, an FIR filter is designed by the Hamming window function. It is implemented as a 21st-order bandpass filter, whose lower and upper cutoff frequencies are 25 Hz and 400 Hz, respectively. Moreover, a 3-order Butterworth filter is implemented with the same cutoff frequencies as the FIR filter.

C. Experimental Setup

We adopt PyTorch (version 2.1.2) and PyTorch Lightning (version 2.1.2) to build our experimental environment, using a server configuration of a Intel(R) Xeon(R) Silver 4214R CPU and a RTX3080Ti GPU. We design a denoising evaluation experiment to judge the denoising ability of the denoisers. Since

TABLE II
5-FOLD CROSS VALIDATION RESULTS FOR RESNET34, VGG11_BN, AND GOOGLNET WITH DIFFERENT DENOISING METHODS.

Classifier	Denoiser	Acc (%)	F1 (%)	Se (%)	Sp (%)	UAR (%)	P-value
ResNet34	DWT	94.6(±0.1)	86.9(±0.1)	89.9(±1.1)	95.8(±0.4)	92.8(±0.4)	.252
	FIR Filter	94.5(±1.0)	86.7(±2.3)	90.8(±1.7)	95.4(±1.1)	93.1(±1.1)	
	Butterworth Filter	96.7(±0.2)	91.9(±0.5)	93.5(±1.2)	97.5(±0.4)	95.5(±0.5)	
	HT Demucs	96.7(±0.3)	92.0(±0.7)	94.0(±0.5)	97.4(±0.4)	95.7(±0.3)	
VGG11_BN	DWT	93.3(±0.7)	84.1(±1.0)	88.9(±2.9)	94.4(±1.5)	91.6(±0.7)	.031*
	FIR Filter	93.1(±1.6)	83.9(±3.1)	90.0(±1.6)	93.8(±1.9)	91.9(±1.3)	
	Butterworth Filter	92.6(±3.2)	83.4(±6.3)	90.8(±3.2)	93.1(±3.8)	91.9(±2.7)	
	HT Demucs	96.1(±0.6)	90.6(±1.3)	93.5(±1.3)	96.8(±0.5)	95.1(±0.8)	
GoogLeNet	DWT	91.1(±1.2)	80.0(±1.8)	89.0(±3.9)	91.6(±2.2)	90.3(±1.1)	.003**
	FIR Filter	94.2(±0.7)	85.8(±1.4)	88.5(±3.6)	95.6(±1.3)	92.0(±1.3)	
	Butterworth Filter	94.3(±1.4)	86.4(±2.8)	90.0(±0.9)	95.4(±1.7)	92.7(±1.0)	
	HT Demucs	95.1(±1.0)	88.5(±2.1)	93.5(±0.8)	95.5(±1.4)	94.5(±0.5)	

1. Data are presented as median ± standard deviation (SD).

2. P-values are calculated for 5-fold UAR with one-tailed paired t-test and 95 % confidence level **between Butterworth Filter and HT Demucs**. Statistically significant differences ($p < .05$) are marked with *, while extremely significant differences ($p < .01$) are marked with **.

3. Before the t-test, we have performed Shapiro-Wilk normality test to guarantee that the data meet the normal distribution (all the p-values $> .2$).

it is hard to obtain the clean PCG signal, we synthesise 10 000 “standard” PCG signals and add White Gaussian Noises to them following the method proposed by Almasi *et al.* [26]. Then, we select 2 000 of them as the test set, and the remaining ones are used to train the HT Demucs with the Adam optimiser, 0.001 as learning rate, and the L1 Loss. Afterwards, we evaluate these denoisers on the test set. In the classification experiment, we first implement the segmentation stage above all, obtaining 81 553 heart sound segments lasting 2.5 s. These segments will be the whole dataset for the remaining stages. After that, the segmented recordings are resampled to 1 kHz to reduce the computing overhead. Then, we divide these segments into 80 % training set and 20 % test set by random sampling¹. The training set is then partitioned into a 5-fold cross-validation, and each fold contains samples of “a-f” subset labels using stratified sampling on 6 subsets. Further, a full training cycle for each iteration carries out 50 epochs, while the validating set (the leave-out set in each iteration) is used for early stopping. The experimental results are calculated for each fold, respectively. In this work, we train each denoiser (or denoising method) with each pretrained classifier in a similar process, which is optimised by the Adam optimiser with a learning rate of 0.001 and the Cross Entropy Loss. The pretrained HT Demucs has 4 output channels, but we only use the first of them (represented as “drum” channel). In addition, the way we transform audios between single channel and dual channel is simply duplicating and averaging.

D. Evaluation Metrics

The denoising ability test is evaluated by signal-to-noise ratio (SNR) and mean squared error (MSE).

The model’s classification performance is evaluated by learning performance indicators including accuracy (Acc), F1-score (F1), sensitivity (Se), specificity (Sp), and unweighted average recall (UAR):

¹Pre-processed dataset for each fold in the experiment can be accessed from <https://github.com/Aaron-Gp/HeartSoundDenoising>.

$$UAR = \frac{\sum_{i=1}^{N_c} Recall_i}{N_c}, \quad (3)$$

where N_c indicates the number of classes.

E. Results

TABLE III
RESULTS FOR DENOISING ABILITY TESTING.

Denoiser	SNR ↑	MSE ↓
DWT	50.31	1.995e-7
FIR Filter	29.38	2.474e-5
Butterworth Filter	11.32	1.581e-3
HT Demucs 1 epoch *	26.60	4.693e-5
HT Demucs 30 epochs *	42.89	1.103e-6

* The HT Demucs is fine-tuned for 1 epoch and 30 epochs, respectively.

Table III shows the results of the denoising test. DWT achieves the best performance in eliminating additive white Gaussian noise and the HT Demucs fine-tuned with 30 epochs can obtains similar scores. Even with 1 epoch fine-tuning, the HT Demucs can outperform the Butterworth filter.

The classification’s experimental results are shown in Table II. We validate each fold on the test set and calculate the average metrics’ value and the standard deviation. In this study, the proposed HT Demucs denoiser achieves the highest UAR with each of the classifiers; the Butterworth filter takes the second place. Furthermore, a significant difference ($p < .05$ by one-tailed paired t-test) can be observed when comparing the HT Demucs and the Butterworth filter with GoogLeNet or the VGG11_BN classifier.

IV. DISCUSSION

Our proposed HT Demucs model as a heart sound denoiser not only exceeds the conventional denoising approach, but also performs more robust given its smaller standard deviation (see

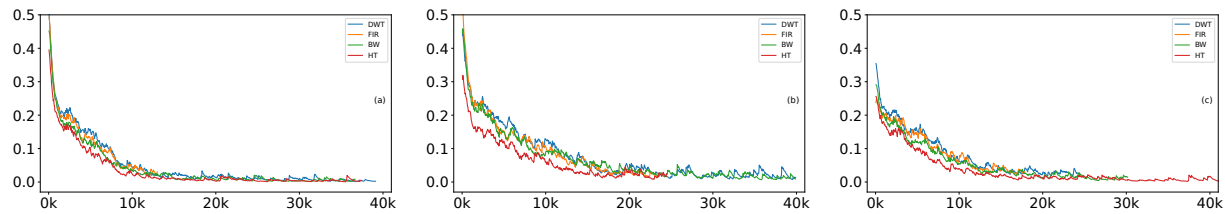


Fig. 3. Training loss curve comparison among all denoisers, where the Butterworth filter is denoted as BW and the HT Demucs is denoted as HT. Examples contain results from each classifier: (a) ResNet34, (b) VGG11_BN, (c) GoogLeNet.

Table II). The ability of each classifier also accounts for the final scores, which is why the differences between each denoiser are not similar with varying classifiers. Yet, they form a consistent ranking that the HT Demucs outperforms the Butterworth filter, then the FIR filter, and then the DWT. This may reflect the ability to deal with real noisy PCG signals. Meanwhile, we also notice that our proposed denoiser can accelerate the convergence of training loss generally with all classifiers we test. Fig. 3 shows the comparative demonstrations we sample from all training processes. An interesting phenomenon is that the denoising ability test results and the classification experiment's results for the 3 classic denoisers seem to be just the opposite. We argue that DWT and FIR filter will smooth the murmurs, which makes it hard to discriminate the abnormality of the heart sounds – “murmurs are not noises” is also why heart sound denoising is so difficult to tackle and evaluate.

V. CONCLUSION

In this study, we proposed a novel approach to denoise heart sounds using an adaptive music source separation model (HT Demucs). We made a contrasting experiment to investigate and compare our method with the classic denoising techniques, where we found that the proposed HT Demucs denoiser obtained a superior UAR (at most 95.7 %) in classifying normal and abnormal heart sounds evaluated on the PhysioNet/CinC database. We conclude that transferring an audio source separation model to denoise PCG signals can be successful and the adaptive denoising method may be advantageous in heart sound classification tasks. Future work could study designing specific models for heart sound classification scenarios based on the audio source separation architecture.

REFERENCES

- [1] Z. Wang, K. Qian, H. Liu, B. Hu, B. W. Schuller, and Y. Yamamoto, “Exploring interpretable representations for heart sound abnormality detection,” *Biomedical Signal Processing and Control*, vol. 82, p. 104569, 2023.
- [2] W.H.O., “Cardiovascular diseases (cvds),” [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)), 2021.
- [3] J. Yanase and E. Triantaphyllou, “A systematic survey of computer-aided diagnosis in medicine: Past and present developments,” *Expert Systems with Applications*, vol. 138, p. 112821, 2019.
- [4] D. Kumar, “Automatic heart sound analysis for cardiovascular disease assessment,” Ph.D. dissertation, University of Coimbra, 2014.
- [5] L. Zhu, K. Qian, Z. Wang, B. Hu, Y. Yamamoto, and B. W. Schuller, “Heart sound classification based on residual shrinkage networks,” in *Proc. EMBC*, 2022, pp. 4469–4472.
- [6] Z. Ren, Y. Chang, T. Tam Nguyen, Y. Tan, K. Qian, and B. W. Schuller, “A comprehensive survey on heart sound analysis in the deep learning era,” arXiv preprint arXiv:2301.09362, 2023.
- [7] M. H. Asmare, F. Woldehanna, L. Janssens, and B. Vanrumste, “Can heart sound denoising be beneficial in phonocardiogram classification tasks?,” in *Proc. EMBC*, 2021, pp. 354–358.
- [8] L. Hamza Cherif, “Digitals filters in heart sound analysis,” *Journal of Clinical Medicine Research*, vol. 1, pp. 97–108, 2014.
- [9] D. Stoller, S. Ewert, and S. Dixon, “Wave-U-Net: A multi-scale neural network for end-to-end audio source separation,” in *Proc. ISMIR*, 2018, pp. 334–340.
- [10] A. Cohen-Hadria, A. Roebel, and G. Peeters, “Improving singing voice separation using deep u-net and wave-u-net with data augmentation,” in *Proc. EUSIPCO*, 2019, pp. 1–5.
- [11] H.-J. Nam and H.-J. Park, “Pitch mark detection from noisy speech waveform using wave-u-net,” in *Proc. ICASSP*, 2023, pp. 1–5.
- [12] K. K. Thakur, S. Choudhury, S. Ghosh, S. Dash, T. S. Chhabra, I. Ali, R. T. Shankarappa, S. Tiwari, and S. Goyal, “Speech enhancement using open-unmix music source separation architecture,” in *Proc. DELCON*, 2022, pp. 1–6.
- [13] S. Rouard, F. Massa, and A. Défossez, “Hybrid transformers for music source separation,” in *Proc. ICASSP*, 2023.
- [14] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *Proc. ICLR*, 2022.
- [15] D. B. Springer, L. Tarassenko, and G. D. Clifford, “Logistic regression-hmm-based heart sound segmentation,” *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 4, pp. 822–832, 2016.
- [16] F. B. Azam, M. I. Ansari, I. McLane, and T. Hasan, “Heart sound classification considering additive noise and convolutional distortion,” arXiv preprint arXiv:2106.01865, 2021.
- [17] J. Ji, L. Zhu, H. Zhang, K. Qian, K. Xu, Z. Song, B. Hu, B. W. Schuller, and Y. Yamamoto, “Weight light, hear right: Heart sound classification with a low-complexity model,” in *Proc. EUSIPCO*. Lyon, France: IEEE, 2024, pp. 326–330.
- [18] X. Qiu, L. Zhu, Z. Song, Z. Chen, H. Zhang, K. Qian, Y. Zhang, B. Hu, Y. Yamamoto, and B. W. Schuller, “Study selectively: An adaptive knowledge distillation based on a voting network for heart sound classification,” in *Proc. INTERSPEECH*, Kos Island, Greece, 2024, pp. 137–141.
- [19] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, “Going deeper with convolutions,” in *Proc. CVPR*, 2015, pp. 1–9.
- [20] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” arXiv preprint arXiv:1409.1556, 2014.
- [21] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. CVPR*, 2016, pp. 770–778.
- [22] T. Koike, K. Qian, Q. Kong, M. D. Plumbley, B. W. Schuller, and Y. Yamamoto, “Audio for audio is better? An investigation on transfer learning models for heart sound classification,” in *Proc. EMBC*, 2020, pp. 74–77.
- [23] C. Liu, D. Springer, Q. Li, B. Moody, R. A. Juan, F. J. Chorro, F. Castells, J. M. Roig, I. Silva, A. E. Johnson *et al.*, “An open access database for the evaluation of heart sound algorithms,” *Physiological Measurement*, vol. 37, no. 12, pp. 2181–2213, 2016.
- [24] M. N. Ali, E.-S. A. El-Dahshan, and A. H. Yahia, “Denoising of heart sound signals using discrete wavelet transform,” *Circuits, Systems, and Signal Processing*, vol. 36, no. 11, pp. 4482–4497, 2017.
- [25] Y. Ghanbari and M. R. Karami-Mollaei, “A new approach for speech enhancement based on the adaptive thresholding of the wavelet packets,” *Speech Communication*, vol. 48, no. 8, pp. 927–940, 2006.
- [26] A. Almasi, M. B. Shamsollahi, and L. Senhadji, “A dynamical model for generating synthetic phonocardiogram signals,” in *Proc. EMBC*, 2011, pp. 5686–5689.